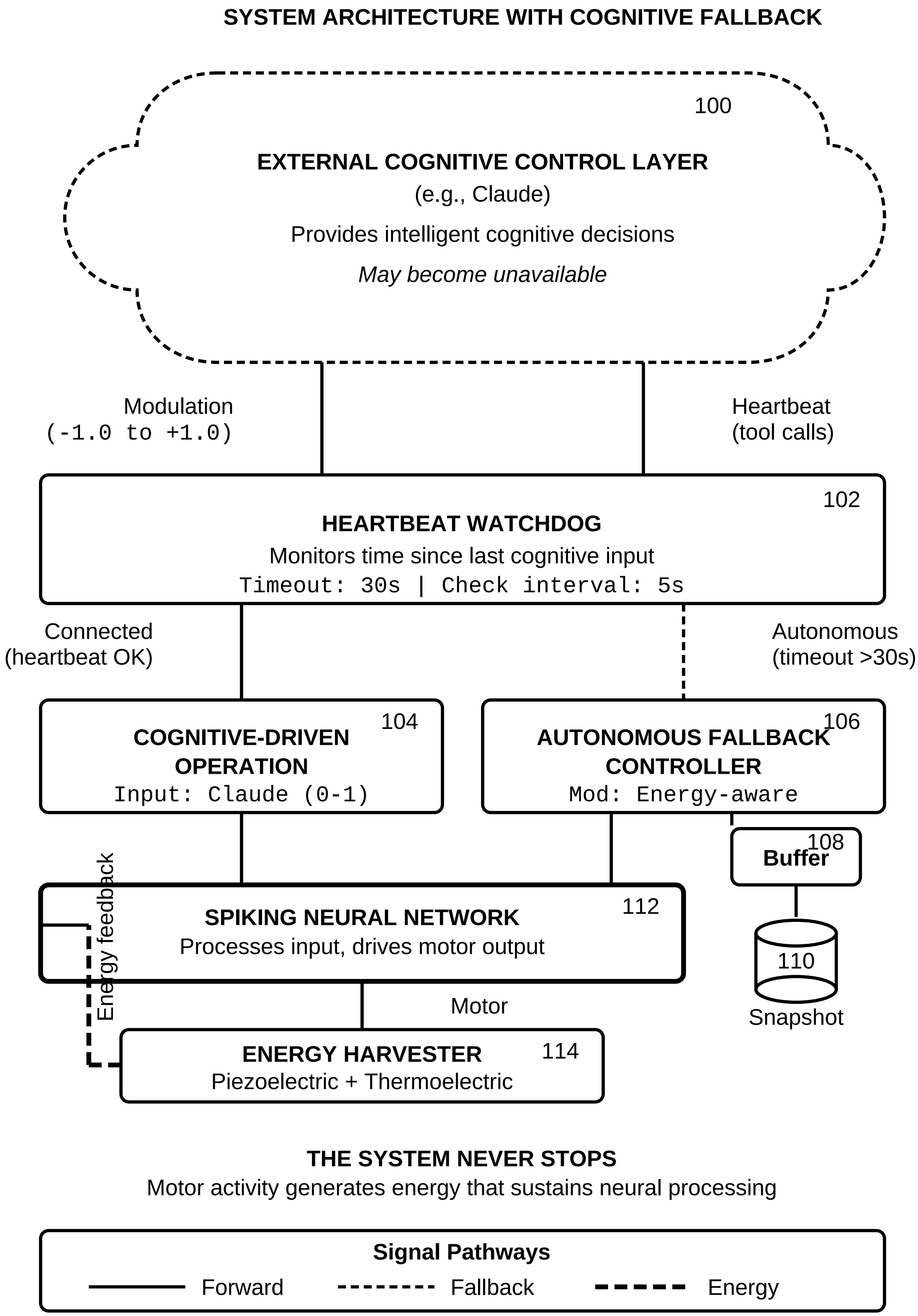
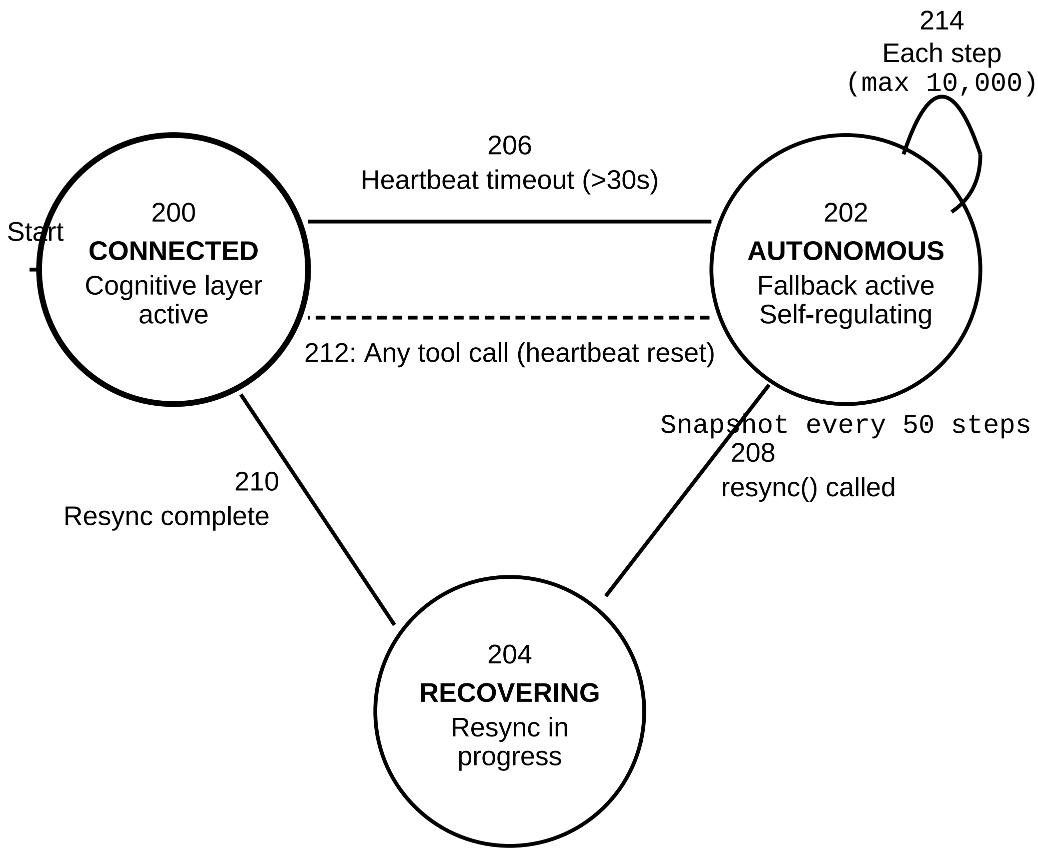

SYSTEM ARCHITECTURE WITH COGNITIVE FALLBACK
100
EXTERNAL COGNITIVE CONTROL LAYER
(e.g., Claude)
Provides intelligent cognitive decisions
May become unavailable
Modulation
(-1.0 to +1.0)
Heartbeat
(tool calls)
102
HEARTBEAT WATCHDOG
Monitors time since last cognitive input
Timeout: 30s | Check interval: 5s
Connected
(heartbeat OK)
Autonomous
(timeout >30s)
104
COGNITIVE-DRIVEN OPERATION
Input: Claude (0-1)
106
AUTONOMOUS FALLBACK CONTROLLER
Mod: Energy-aware
108
Buffer
110
Snapshot
Energy feedback
112
SPIKING NEURAL NETWORK
Processes input, drives motor output
Motor
114
ENERGY HARVESTER
Piezoelectric + Thermoelectric
THE SYSTEM NEVER STOPS
Motor activity generates energy that sustains neural processing
Signal Pathways
Forward
Fallback
Energy

FIG. 1

STATE TRANSITION DIAGRAM



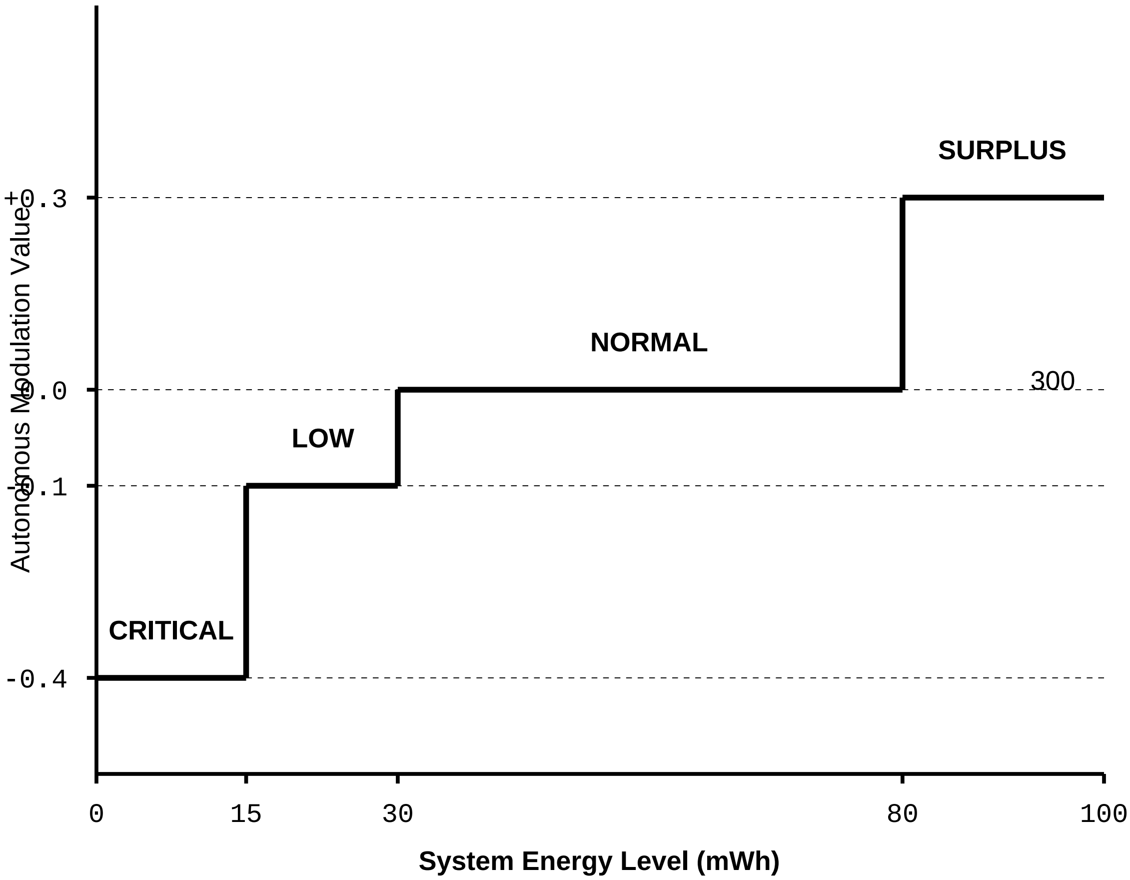
Transition Summary

206: Timeout triggers fallback
208: Explicit resync request
210: Resync completes
212: Direct reconnection
214: Autonomous step loop

—— Initial state
- - - - - - - Direct reconnect

FIG. 2

ENERGY-AWARE MODULATION CURVE



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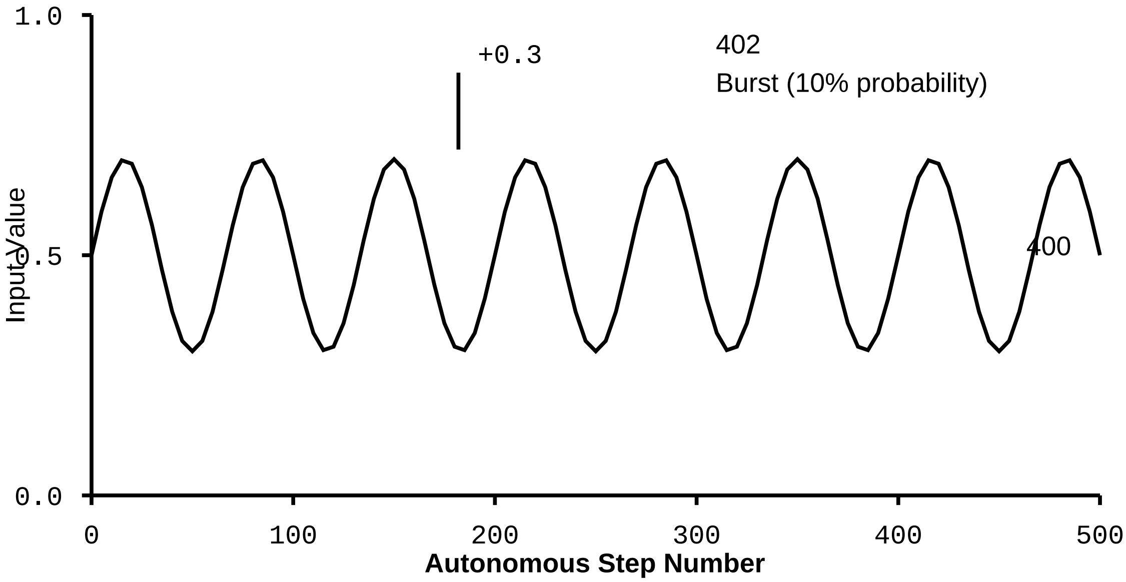
Behavioral Effects

< 15 mWh	mod = -0.4	Maximum conservation
15-30 mWh	mod = -0.1	Cautious operation
30-80 mWh	mod = 0.0	Standard processing
> 80 mWh	mod = +0.3	Exploration enabled

FIG. 3

AUTONOMOUS INPUT GENERATOR OUTPUT



404

Autonomous Input Formula

circadian = 0.5 + 0.2 x sin(2pi x t x 3)

burst = 0.3 (with 10% probability per step)

Energy Recovery During Autonomous Operation

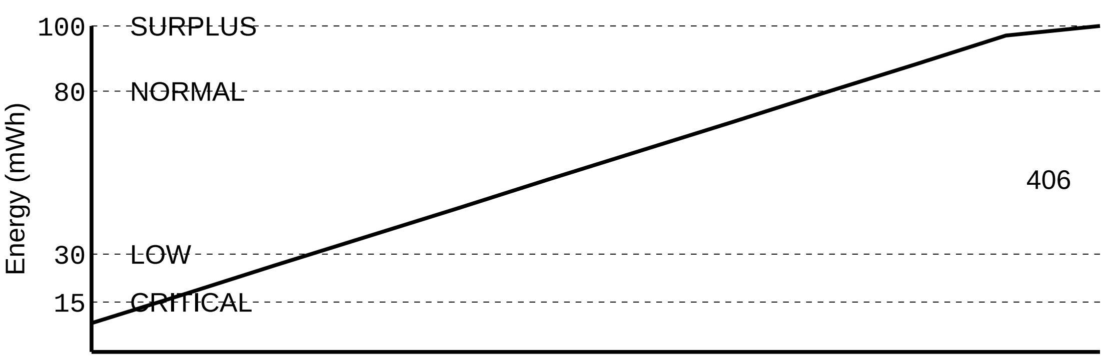


FIG. 4

RESYNCHRONIZATION PAYLOAD STRUCTURE

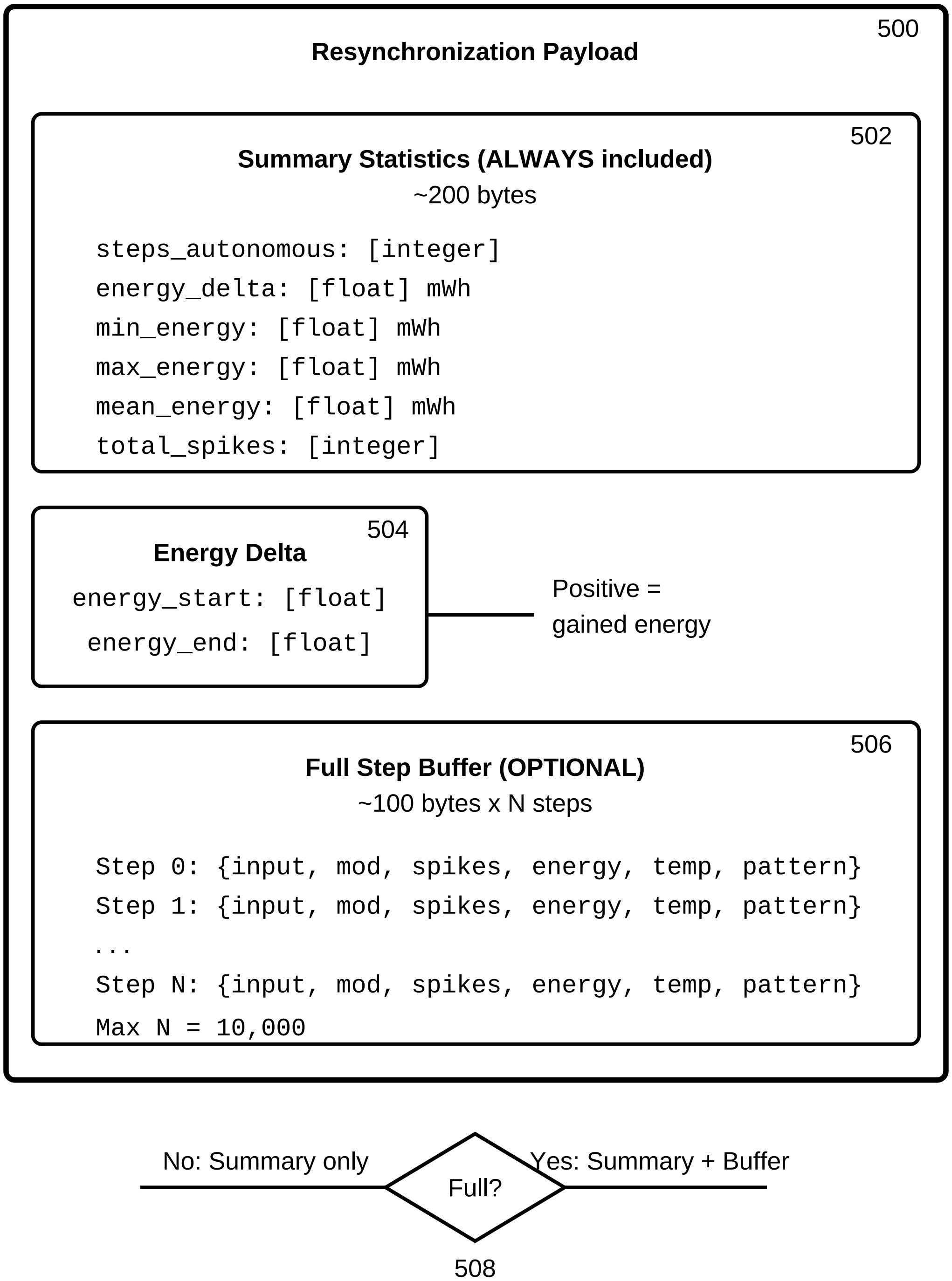
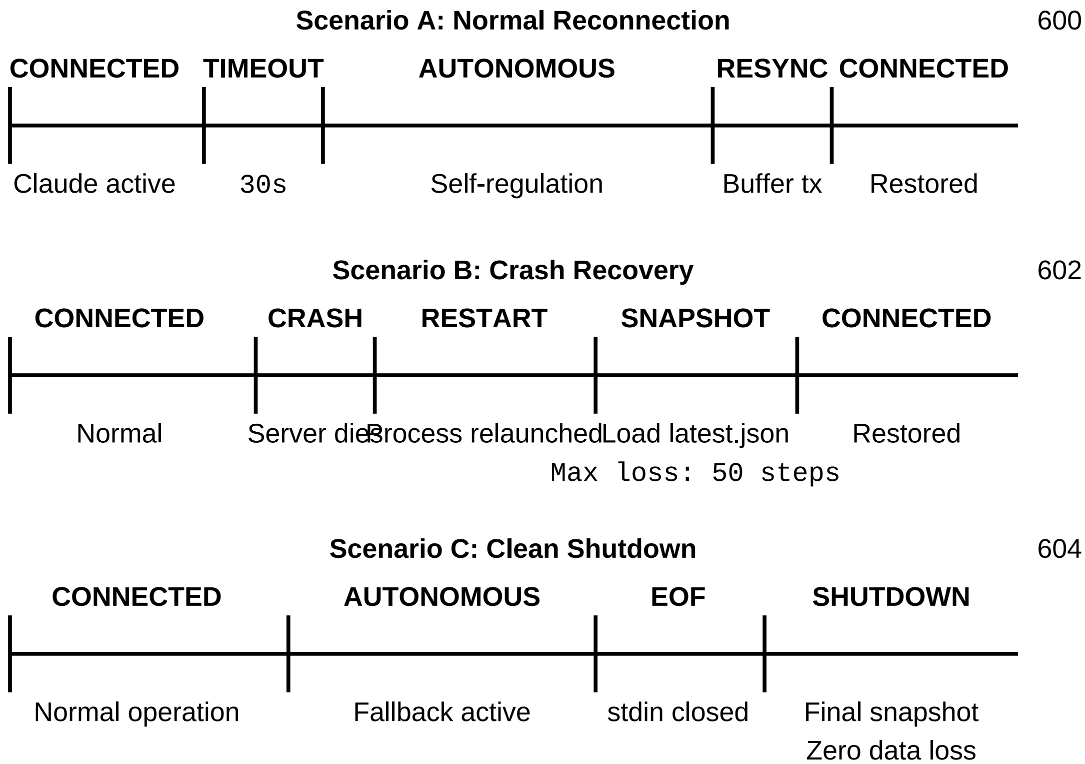


FIG. 5

RECOVERY SCENARIOS



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Recovery Guarantees

Normal reconnection: Full continuity, resync payload transmitted

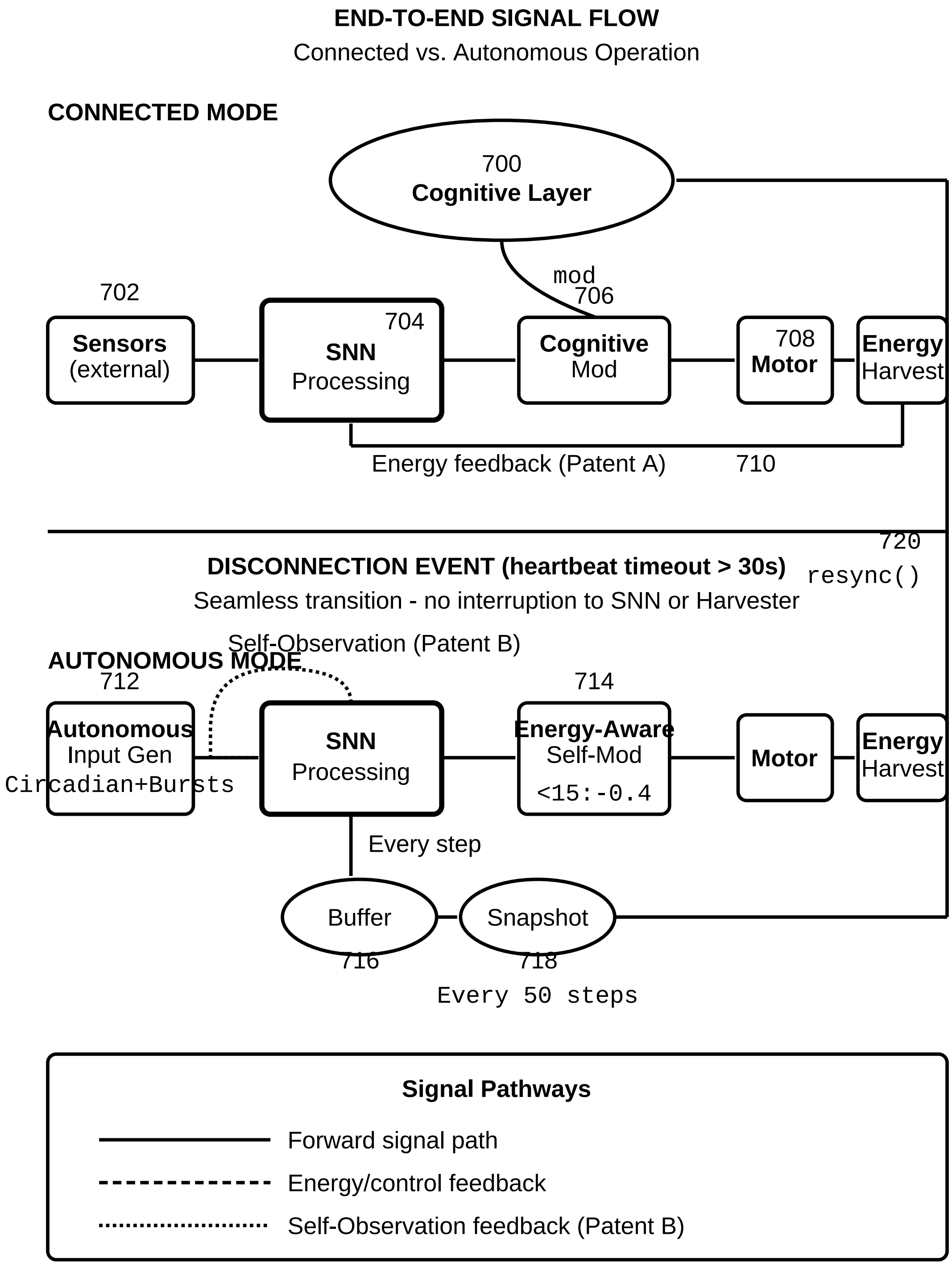
Crash recovery: Max 50 steps lost (snapshot every 50 steps)

Clean shutdown: Zero data loss (final snapshot on EOF)

Max autonomous duration: 10,000 steps (~50 hours)

System actively processing

FIG. 6



END-TO-END SIGNAL FLOW
Connected vs. Autonomous Operation
CONNECTED MODE
700
Cognitive Layer
mod
702
Sensors
(external)
704
SNN
Processing
706
Cognitive
Mod
708
Motor
Energy
Harvest
Energy feedback (Patent A)
710
720
resync()
DISCONNECTION EVENT (heartbeat timeout > 30s)
Seamless transition - no interruption to SNN or Harvester
Self-Observation (Patent B)
AUTONOMOUS MODE
712
Autonomous
Input Gen
Circadian+Bursts
SNN
Processing
714
Energy-Aware
Self-Mod
<15:-0.4
Motor
Energy
Harvest
Every step
Buffer
Snapshot
716
718
Every 50 steps
Signal Pathways
Forward signal path
Energy/control feedback
Self-Observation feedback (Patent B)

FIG. 7